

Project Planning Phase

Project Planning (Product Backlog, Sprint Planning, Stories, Story points)

Date	20 February 2026
Team ID	LTVIP2026TMIDS24635
Project Name	Flight finder-navigating your air travel options
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

The product backlog represents a prioritized list of features and tasks required to complete the project. These tasks are divided into sprints, and each task is assigned story points based on complexity and effort.

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	Collect data from various APIs/sources	2	High	1
Sprint-1	Data Collection	USN-2	Load collected data into storage/memory	1	High	2
Sprint-1	Data Preprocessing	USN-3	Handle missing values in dataset	2	Low	3
Sprint-1	Data Preprocessing	USN-4	Convert/encode categorical data	2	Medium	4
Sprint-2	Model Building	USN-5	Build a machine learning model to recommend or classify flights	5	High	1
Sprint-2	Model Building	USN-6	Test model accuracy and performance	3	High	2
Sprint-2	Deployment	USN-7	Design working HTML pages for frontend	3	Medium	3
Sprint-2	Deployment	USN-8	Deploy using Flask and integrate model	5	High	4

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	8	5 Days	4 February 2026	10 February 2026	8	10 February 2026
Sprint-2	16	5 Days	11 February 2026	18 February 2026	16	18 February 2026

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\text{sprint duration}}{\text{velocity}} = \frac{20}{10} = 2$$

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

